

Tatevik Mkrtchyan

PhD Student in Astrophysics, Universidad Diego Portales, Santiago, Chile
tatevik.mkrtchyan@mail.udp.cl — dathevik.com

RESEARCH INTERESTS

Active Galactic Nuclei, High-redshift quasars, Galaxy Evolution and Formation, Supermassive Black Holes, Photometric Surveys

EDUCATION

PhD Student in Astrophysics

Instituto de Estudios Astrofísicos, Universidad Diego Portales
Thesis Title: Characterizing Active Galactic Nuclei across Cosmic Time

August 2022 — Present
Santiago, Chile

Master of Science in Astrophysics

Faculty of Physics, Yerevan State University
Thesis Title: The Study of High Energy Gamma-Ray Emission from Galaxy Clusters

September 2019 — May 2021
Yerevan, Armenia

Bachelor of Science in Physics

Faculty of Physics, Yerevan State University
Thesis Title: High Energy Gamma Ray Radiation Processes of Blazars

September 2014 — June 2019
Yerevan, Armenia

ACADEMIC EXPERIENCE

Senior Lab Assistant

Byurakan Astrophysical Observatory

April 2021 — September 2022
Byurakan, Armenia

- Worked on Galaxy Evolution models using Principal Component Analysis.

Junior Researcher Intern

International Center for Relativistic Astrophysics

December 2016 — March 2018
Yerevan, Armenia

- Made models of high-energy gamma-ray radiation processes of blazars using Wolfram Mathematica.

PUBLICATIONS

First Author

- A MUSE Search for Extended Lyman-alpha Halos around High-Redshift Radio-Loud quasars
T. Mkrtchyan, C. Mazzucchelli, E. P. Farina, et al. (in prep.)
- 4MOST ChANGES: Catalog of high-redshift quasar candidates ($4.5 < z < 7$) selected with SED fitting
T. Mkrtchyan, C. Mazzucchelli, R. J. Assef, M. J. Temple, et al. 2026, A&A
- Inclusive Teaching Methods Handbook: Published by The Republican Pedagogical-Psychological Center of RA

For a complete list of publications, visit: [ADS Publication Profile](#)

OBSERVATIONS

Observing run with EFOSC2 instrument in the NTT

La Silla Observatory, 4 nights

La Serena, Chile
January 2024

Observing run with FIRE instrument in the 6.5-meter Baade Magellan Telescope

Las Campanas Observatory, 2 nights

La Serena, Chile
August 2023

Observing run with FIRE instrument in the 6.5-meter Baade Magellan Telescope

Las Campanas Observatory, 1 night

La Serena, Chile
May 2023

Observing run with FIRE instrument in the 6.5-meter Baade Magellan Telescope

Las Campanas Observatory, 2 nights

La Serena, Chile
April 2023

CONFERENCES AND PRESENTATIONS

Conferences

The Baryon Cycle from Reionization to Cosmic Noon [Contributed Talk]	Dec 2025
South American Center for Astronomy, Chinese Academy of Sciences	Puerto Varas, Chile
Highly Accreting Supermassive Black Holes Across all Cosmic Times [Contributed Talk]	Dec 2025
European Southern Observatory	Santiago, Chile
SOCHIAS XX Annual Meeting [Contributed Talk & Poster]	Oct 2025
SOCHIAS, Universidad Austral de Chile	Puerto Montt, Chile
Extreme Galaxies in their Extreme Environments at Extremely Early Epochs [Contributed Poster]	May 2024
The National Institute for Astrophysics	Reykjavik, Iceland
The Santiago AGN Workshop: A Holistic View of Active Galaxies [Contributed Talk]	Nov 2023
European Southern Observatory	Santiago, Chile
New Era of AGN Science with the Vera C. Rubin LSST [Contributed Talk]	Jul 2023
LSST AGN Science Collaboration, National Radio Astronomy Observatory	Charlottesville, USA

Seminars

Weekly Seminar	Jun 2026
The Alikhanyan National Science Laboratory, Center for Cosmology and Astrophysics	Yerevan, Armenia
Weekly Seminar	Feb 2026
European Southern Observatory, Extragalactic Group	Santiago, Chile
Astronomy Tuesday Talk	Nov 2025
University of California, Los Angeles	Los Angeles, USA
Astronomy Tea Talk	Nov 2025
Cahill Center for Astronomy and Astrophysics, California Institute of Technology	Los Angeles, USA
Gemini North Observatory Talk	Nov 2025
NSF NOIRLab	Hilo, USA

Schools

18th Heidelberg Summer School [Participant]	Sep 2023
Max Planck Institute for Astronomy, University of Heidelberg	Heidelberg, Germany
Science Leadership Conference [Participant]	Sep 2020
The Optical Society (Optica)	Remote
Sixth Byurakan Summer School [Participant]	Sep 2018
Byurakan Astrophysical Observatory	Byurakan, Armenia
Wolfram Summer School [Participant]	Aug 2017
Wolfram Research Company	Yerevan, Armenia

NON-ACADEMIC EXPERIENCE

Education Expert/Lead Physics Teacher	2020 — 2022
National Center of Educational Technologies	Yerevan, Armenia
- Mentored teachers in implementing remote teaching methods, developed quality assurance tools for educational programs. - Made 100+ videos based on a physics school program available on YouTube.	
Teacher Trainer/Mentor	2019 — 2022
State Pedagogical-Psychological Center	Yerevan, Armenia
Mentored educators in effective classroom management techniques and wrote a handbook on STEM teaching methods.	
Physics/Astrophysics Teacher	2018 — 2019
Shirakatsy Lyceum International Scientific-Educational Complex	Armenia
Led “Young Astrophysicist” student club and mentored students in scientific research projects.	
Robotics Workshop Leader	2017 — 2018
TUMO Center for Creative Technologies	Armenia
Taught robotics to children aged 12-18 and developed STEM curriculum.	

SKILLS

- **Languages:** Armenian (Native), Russian (Fluent), English (Fluent), Spanish (Intermediate)
- **Technical:** Python (matplotlib, pandas, astropy, scipy), Wolfram Mathematica, LePhare, EsoReX, LaTeX, Git
- **Soft Skills:** Scientific Communication, Teaching, Research Leadership, Project Management

AWARDS

Beca Adelina Gutiérrez Sociedad Chilena de Astronomía	Chile 2024
Finalist of Falling Walls Science Engage International science communication competition	Global 2020
Scholarship after Viktor Hambardzumyan Prestigious astronomy scholarship	Armenia 2020
Scholarship after Gevorg and Gohar Vardanian Academic excellence scholarship	Armenia 2014
Annual Presidential Award of the RA in IT Sphere Recognition for contributions to information technology	Armenia 2014

REFERENCES

Dr. Chiara Mazzucchelli *Supervisor, Instituto de Estudios Astrofísicos, Universidad Diego Portales, Santiago, Chile*
E-mail: chiara.mazzucchelli@mail.udp.cl

Dr. Emanuele Farina *Collaborator, International Gemini Observatory, Hilo, USA*
E-mail: emanuele.farina@noirlab.edu

Additional references available upon request